Part 1 of 2

MAKING PCBs AT HOME Photographically

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Here is a comprehensive guide to make your own PCBs at home using materials that are readily available in shops and online stores. While it addresses single-sided PCBs, it can be extended to create double-sided PCBs as well. The steps involved in making single-sided PCBs, in that order, are:

1. Create a schematic
2. Create a PCB layout
3. Create the artwork
4. Transfer the artwork onto the PCB laminate; let's call this stage 'artwork transfer' or 'exposing the PCB'
5. Mask the area where you want to retain copper. This stage is called 'developing'
6. Remove copper from the PCB laminate where you don't want it, e.g., tracks, etc. This process is called 'etching'
7. Remove the masking material to reveal the PCB
8. Drill holes to solder and mount components and accessories or sub-assemblies. This stage may not be required if the PCB is being prepared for surface-mount components
9. Tin the copper tracks to avoid oxidation of copper
10. Deposit some sort of protective material on the side of the PCB that contains copper tracks, also called 'solder side.' This material does not allow solder to stick to the copper tracks other than where solder needs to stick, like component pads. This process is called 'masking'
11. Print the 'silk screen' on the components side of the PCB to mark the component references corresponding to the components in the schematic

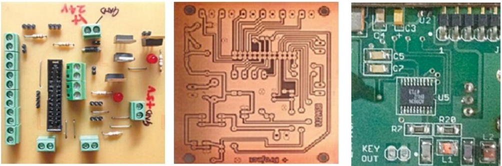
This article details steps 1 through 8 as these are sufficient for most applications and can be performed at home.

However, your project circumstances and requirements will determine the steps that you need to perform. For example, if the artwork is already available, you may skip steps 1 and 2 and jump to step 3.

Step 1: Create a schematic

The prerequisite for this step is that the circuit for which you want to create the schematic is available along with component details like values, power rating and package details (size of the component, the spacing and layout of its connecting pins, etc). If that's not the case, you may consider using some kind of simulation software like LT Spice or Proteus to draw the schematic and perform simulation on it. This may save you both time and money as:

1. Experimenting and reworking on a software simulated model is far less time consuming than procuring all the components that would be required for designing your circuit
2. You need not buy the components till you simulate and check everything theoretically to a large extent on the software.



(a) Components-side view of a finshed homebrewed PCB using through-hole components

(b) Copper-side or solder-side view of a homebrewed PCB using through-hole components

(c) A PCB using surface-mounted components

Fig. 1: Finished home-brewed PCBs